Program : **Diploma in Civil and Environmental Engineering**	
Course Code : **4352**	Course Title: **Environmental Chemistry and Microbiology**
Semester :**4**	Credits: **4**
Course Category: **Programme Core Course**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- Understand concepts of physical chemistry significant to environmental engineering.
- Understand chemical characteristics of organic pollutants and colloids significant to environmental engineering.
- Understand the principles of determination of water quality parameters and soil parameters.
- Develop understanding of characteristics, classification and detection of microorganism and their interaction with environment.
- Apply principles of microbiology in understanding and solving environmental problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of chemistry		Secondary school	
Solutions, pH, pOH, buffer solutions, volumetric analysis	1004	Applied Chemistry	1
Characteristics of potable water and brief idea on treatment steps	1004	Applied Chemistry	1
Water pollution- Sources, important water quality parameters, pH, DO, BOD, COD	2001	Environmental Science	2
Air pollution- Types of pollutants and control measures	2001	Environmental Science	2
Bio geo cycles- phosphorus cycle, carbon cycle and nitrogen cycle.	2001	Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	To understand fundamentals of general chemistry, physical chemistry and quantitative chemistry significant to environmental engineering.	15	Understanding
CO2	Understand properties of organic compounds and colloids significant to environmental engineering and apply physical chemistry to solve environmental issues.	14	Applying
CO3	Understand characteristics of various microorganisms significant to environmental engineering.	15	Understanding
CO4	Apply principles of microbiology in pollution mitigation.	14	Applying
	Tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3	2		
CO2	3				2		
CO3	3			3	2		
CO4	3				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To understand fundamentals of general chemistry, physical chemistry and quantitative chemistry significant to environmental engineering.**		
M1.01	Understand significance of chemistry in environmental engineering	1	Understanding
M1.02	Understand gas laws and their application in environmental engineering	2	Understanding

M1.03	Understand state of equilibrium and Le Chatlier’s principle.	3	Understanding
M1.04	Understand basic concepts of quantitative chemistry and instrumental methods of analysis.	9	Understanding
	Total	**15**	

Contents:

Significance of chemistry in environmental Engineering.

Gas laws and their application in environmental engineering- Boyle’s law, Charles law, Dalton’s law of partial pressure, Henry’s law, Graham’s law, Gay-Lussac’s law of combining volumes.

Chemical equilibrium -Le Chatlier’s principle, Solubility product, ways of shifting chemical equilibria -brief description and application in environmental engineering.

Basic concepts of quantitative chemistry and instrumental methods of analysis- Expression of results -ppm and SI units, Types of samples in water analysis, Good laboratory practices, selection of laboratory apparatus and reagents. Precipitation, Filtration, drying or ignition and dessication. Analytical balance.

Fundamentals of gravimetric analysis, concepts and techniques of volumetric analysis-calibrated glassware, equivalent or normal solutions, primary standards, secondary standards, major type of reactions in volumetric analysis- Acid base titration, precipitation methods, oxidation reduction methods, applications in water and soil analysis.

Colorimetry- Lambert’s law, Beer’s law, spectrophotometers, applications in water and soil analysis.

Basic principles of turbidimetry, nephlometry and chromatography and their applications in environmental engineering.

CO2	**Understand properties of organic compounds and colloids significant to environmental engineering and apply physical chemistry to solve environmental issues**.		
M2.01	Understand environmental significance of organic compounds.	2	Understanding
M2.02	Understand characteristics and environmental significance of colloids.	4	Understanding
M2.03	Apply principles of physical chemistry for removal of contaminants.	8	Applying
	Series 1	**1**	

<table>
<tr><td></td><td>Total</td><td>15</td><td></td></tr>
<tr><td colspan="4">Contents
Organic compounds- Properties, Sources, Environmental significance and toxic effect of organic compounds-hydrocarbons, pesticides, and detergents.
Colloidal chemistry-Size, methods of formation, general properties, and environmental significance of colloids, colloidal dispersion in liquids, colloidal dispersion in air.
Basic concepts of chemical kinetics- order and molecularity.
Basic principles, process and environmental applications of coagulation, osmosis, reverse osmosis, electro dialysis, ion exchange, Adsorption, solvent extraction.</td></tr>
<tr><td>CO3</td><td colspan="3">Understand characteristics of various microorganisms significant to environmental engineering.</td></tr>
<tr><td>M3.01</td><td>Understand concept and significance of environmental microbiology</td><td>1</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Understand characteristics of different microorganisms</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Understand bacterial growth and metabolism</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>Understand different microbial environments and interaction of microbes with environment</td><td>3</td><td>Understanding</td></tr>
<tr><td>M3.05</td><td>Understand steps in bacteriological analysis of water and waste water.</td><td>5</td><td>Understanding</td></tr>
<tr><td></td><td>Total</td><td>15</td><td></td></tr>
<tr><td colspan="4">Contents:
Introduction to environmental microbiology- Concept, applications and scope.
Classification of microorganism and their characteristics-bacteria, fungi, slime molds, protozoa, algae, virus.
Bacterial growth and metabolism- Different phases of growth, effect of substrate concentration on growth, aerobic and anaerobic metabolism.
Microbiology of water, waste water, soil and air, water borne diseases, Indicator microbes
Bacteriological analysis of water and waste water- Sampling and processing of water and waste water, cultural methods, MPN index and Membrane filtration techniques for coliforms.</td></tr>
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CO4	**Apply principles of microbiology in pollution mitigation.**		
M4.01	Apply principles of microbiology for treatment of water and waste water.	4	Applying
M4.02	Understand process of bioremediation.	3	Understanding
M4.03	Apply principles of microbiology in remediation of soil and water contaminated with metals	4	Applying
M4.04	Understand application of microbiology in air pollution mitigation	2	Understanding
M4.05	Understand environmental applications of genetically modified organisms.	1	Understanding
	Series 2	**1**	
	Total	**15**	

Contents:

Applications of microbiology in sanitary engineering-Disinfection of water- principle of chlorination, and chlorine demand. Brief description on role of microorganisms waste water engineering-BOD, aerobic biological oxidation, biological nitrification, biological denitrification, biological phosphorus removal, anaerobic fermentation and oxidation (brief description of microbiology and applications, detailed process description not required.)

Bioremediation-Process of biodegradation and methods of bioremediation.

Remediation of metal contaminated soil and water using microbes- principle and process.

Microbiology applied to air pollution mitigation- bio scrubbers and bio filters.

Genetically modified organisms for environmental applications.

Text / Reference:

T/R	Book Title/Author
T	Clair N Sawyer, Perry L McCarty, Gene F Parkin- Chemistry for Environmental Engineering and Science- Mc Graw Hill
T	Ian L Pepper, Charles P Gerba, Terry J Gentry, Environmental Microbiology, Academic Press
R	P D Sharma- Environmental microbiology-Rastogi Publications
R	Samir K Banerji- Environmental Chemistry, PHI Learning Private Limited
R	Rao C S, Environmental Pollution Control Engineering, New Age International

	Limited, Publishers
R	Gary W vanLoon and Stephen J Duffy- Environmental Chemistry-A global perspective, Oxford University press
R	Wastewater Engineering- Treatment and Reuse- Metcalf and Eddy- Tata McGraw Hill
R	S.K Banerjee- Environmental Chemistry, PHI Learning
R	F.W.Fifield and P J Haines -Environmental Analytical Chemistry- Blackwell Science
R	McKinney R E-Microbiology for Sanitary Engineers-McGraw Hill

Online resources:

Sl.No.	Website Link
1	https://youtu.be/Ym3HyZGsOg4?si=HiRav7jRJ69sPknC (Environmental Chemistry-IIT Roorkee)
2	mod01lec01 (youtube.com) (Lecture series on Applied Environmental Microbiology by IIT Roorkee